

# Lab Experiments 8-10

Data Handling & Visualization: Python Code & Output

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## Experiment 8

### Five Most Sold Products

#### PYTHON

##### Code

```
import pandas as pd

df = pd.read_csv("Sales_data.csv")
top_5_products = df.groupby("Product_Name")["Price"].sum().nlargest(5)

print("Five most sold products: ", top_5_products)
```

##### Output

```
Five most sold products:  Product_Name
Widget C      2947.47
Widget H      2748.86
Widget G      2298.22
Widget D      2193.02
Widget F      2007.40
Name: Price, dtype: float64
```

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## Experiment 9

### House Data Analysis

#### PYTHON

##### Code

```
import pandas as pd

df = pd.read_csv("house.csv")

location_price_average = df.groupby("Location")["price"].mean()
more_than_4beds = df[df["beds"] > 4].count()["beds"]
largest_sq_feet = df["size"].max()

print("Average house price for each location: ", location_price_average)
print("Number of houses with more than 4 bedrooms: ", more_than_4beds)
print("Largest house size: ", largest_sq_feet)
```

##### Output

```
Average house price for each location:  Location
Downtown      499218.448333
Hillside      494531.001667
Riverside     508303.572500
Suburbs       433748.390000
```

Uptown            511786.931429  
Name: price, dtype: float64  
Number of houses with more than 4 bedrooms: 7  
Largest house size: 4128

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## Experiment 10

# Monthly Sales Visualization

## PYTHON

### Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('Sales_data.csv')
monthly_sales = df.groupby("Month_sales")["Price"].sum()

plt.plot(monthly_sales.index, monthly_sales.values, marker='o')
plt.title("Monthly Sales Data")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=60)
plt.show()

plt.bar(monthly_sales.index, monthly_sales.values)
plt.title("Monthly Sales Data")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=60)
plt.show()
```

### Output

